Combinations vs. Permutations: Subsets & Prime Cancellation

Probability, Random Variables & Combinatorics • Standard: CCSS.MATH.HSS.CP.B.9

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1. MATHEMATICAL CONCEPT & STACK APPROACH: Calculate permutations and combinations by canceling common factorial terms on paper and verifying on the stack.

Combinations nCr = n! / (r!(n-r)!) count unordered selections by dividing ordered permutations by the r! duplicate arrangements of the chosen group. Canceling 5! from 8! leaves (8 × 7 × 6) / 3! = 56.

2. MODELING WITH BLOCKS (WHEN APPLICABLE) & STUDENT PREDICTION

Published Block Kit: Fraction Studio (84 mm Probability Towers) & Prime Factor Studio

Hands-On Setup: Use Prime Factor blocks to model prime factor cancellation in 8! / (3! × 5!) = (8 × 7 × 6) / (3 × 2 × 1) = 8 × 7 = 56.

Student Prediction (Before Calculator): How many ways to choose a committee of 3 from 8 candidates? 8C3 = 8! / (3! × 5!) = (8 × 7 × 6) / 6 = 56.

3. CALCULATOR EXECUTION (SC-32 KEYCAPS & 4-LEVEL STACK TRACE)

Keystroke Flow: 8 ↶ x! 3 ↶ x! ÷ 5 ↶ x! ÷

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
8	8	Total candidates n = 8
↶ x!	40320	8! = 40,320 in register X
3	3	Subset r = 3 (40320 lifts to Y)
↶ x!	6	3! = 6 in X
÷	6720	40320 / 6 = 6720 in X
5	5	Remaining n - r = 5 (6720 lifts to Y)
↶ x!	120	5! = 120 in X
÷	56	6720 / 120 = 56 combinations!

4. MATHEMATICAL CONTEXT & REAL-WORLD MOTIVATION (Sir Francis Galton's 1873 Bean Machine & The Mathematics of Small Samples)

In 1873, Sir Francis Galton built a vertical board studded with pins (the Quincunx) to show how balls bouncing left or right in combinations nCr settle into a bell-shaped curve—while in 1908 in Dublin, chemist William Sealy Gosset ('Student') developed small-sample statistics with n − 1 degrees of freedom! Students cancel factorial primes with Prime Factor blocks and model exact probability ratios.

Core Investigation (Student Warm-Up A):

A lottery draws 4 winning numbers from a pool of 10 (order does not matter). Simplify 10C4 = 10! / (4! × 6!) by canceling 6! first, then check your result.

Sample Mean & Variance of Discrete Data Sets

One-Variable Data Analysis & Summations • Standard: CCSS.MATH.HSS.ID.A.2

1. FOLLOW-UP EXTENSION CONCEPT, BLOCKS & PREDICTION

Concept & Stack Approach: Statistical measures of center (Mean x-bar) and spread (Standard Deviation s) summarize an entire distribution of values. Finding the sum of observations (Σx) and dividing by the sample count n locates the balance point of the data set.

Blocks & Prediction: Lay out the 5 sample values (4, 6, 7, 8, 10) with Digit Blocks on the Place Value Tray to see how distances above and below 7 balance. Prediction: Sum = 4 + 6 + 7 + 8 + 10 = 35. Sample size n = 5. Mean x-bar = 35 / 5 = 7.0.

2. EXTENSION CALCULATOR SEQUENCE & STACK TRACE

[4] [Σ+] [6] [Σ+] [7] [Σ+] [8] [Σ+] [1] [0] [Σ+]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
[4] [Σ+]	1	First score entered: sample size n = 1
[6] [Σ+]	2	Second score: n = 2
[7] [Σ+]	3	Third score: n = 3
[8] [Σ+]	4	Fourth score: n = 4
[1] [0] [Σ+]	5	Fifth score: n = 5 total data points

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Forgetting to clear the statistical registers ([CLΣ]) before entering a new dataset with [Σ+].

Hands-On Block Remediation: Notice that pressing [Σ+] displays the sample count n in Register X; if the first data point shows n > 1, press [CLΣ] first so prior data does not corrupt the mean.

Misconception 2: Dividing by r! and then multiplying by (n − r)! when computing C(n, r) = n! / (r!(n − r)!).

Hands-On Block Remediation: Compute n! into Y, compute r! and (n − r)! and multiply them together in X ([×]), then press [÷] once to divide n! by the combined denominator.

Extension Challenge (Student Warm-Up B):

Given the four data points {12, 15, 18, 21}, find the sample size n, the total sum of the values, and the sample mean x-bar.

Expected Answer: Count n = 4, Sum = 66, Mean x-bar = 66 / 4 = 16.5.

Combinatorics, Probability Ratios & Series Accumulation

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: COMBINATIONS, PERMUTATIONS & SAMPLE STATISTICS RULES

- Factorial n! = n × (n − 1) × ... × 1 counts the ways to arrange n distinct items; Prime Factor blocks show how common factors cancel in n! / r!.
- Permutations (order matters): nPr = n! / (n − r)! counts ordered arrangements of r items chosen from n.
- Combinations (order does not matter): nCr = n! / (r! × (n − r)!) divides out the r! duplicate orderings.

Key Sequence: 8 ↶ x! 3 ↶ x! ÷ 5 ↶ x! ÷

Investigation 1: Evaluate 6! (the number of ways to arrange 6 distinct books on a shelf).

1. Slot Diagram / Data Sketch:

2. Factorial / Statistical Steps:

Formula (nPr, nCr, or Stat):

Factorial / Sum Expansion:

Computed Value:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 2: How many ways can 5 runners finish in 1st, 2nd, and 3rd place? (Compute 5P3 = 5! / 2!).

1. Slot Diagram / Data Sketch:

2. Factorial / Statistical Steps:

Formula (nPr, nCr, or Stat):

Factorial / Sum Expansion:

Computed Value:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 3: How many 2-person partnerships can be formed from 7 students? (Compute 7C2 = 7! / (2! × 5!)).

1. Slot Diagram / Data Sketch:

2. Factorial / Statistical Steps:

Formula (nPr, nCr, or Stat):

Factorial / Sum Expansion:

Computed Value:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 4: Find the mean x-bar of the two data values 5 and 15.

1. Slot Diagram / Data Sketch:

2. Factorial / Statistical Steps:

Formula (nPr, nCr, or Stat):

Factorial / Sum Expansion:

Computed Value:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Unit 11: Combinatorics, Probability Ratios & Series Accumulation — Answer Key

Complete solutions and SC-32 verification keystrokes for Grades 11–12 / AP

1. CORE (WARM-UP A) & EXTENSION (WARM-UP B) SOLUTIONS

Core Investigation (Warm-Up A): A lottery draws 4 winning numbers from a pool of 10 (order does not matter). Simplify 10C4 = 10! / (4! × 6!) by canceling 6! first, then check your result.

Core Solution: (10 × 9 × 8 × 7) / (4 × 3 × 2 × 1) = 5040 / 24 = 210 combinations.

Core Check Keys: [1] [0] [↶] [x!] [4] [↶] [x!] [÷] [6] [↶] [x!] [÷]

Extension Challenge (Warm-Up B): Given the four data points {12, 15, 18, 21}, find the sample size n, the total sum of the values, and the sample mean x-bar.

Extension Solution: Count n = 4, Sum = 66, Mean x-bar = 66 / 4 = 16.5.

Ext. Check Keys: [1] [2] [Σ+] [1] [5] [Σ+] [1] [8] [Σ+] [2] [1] [Σ+]

2. STUDENT LAB WORKSHEET SOLUTIONS (INVESTIGATIONS 1–4)

Investigation 1: Evaluate 6! (the number of ways to arrange 6 distinct books on a shelf).

Expected Answer: 6 × 5 × 4 × 3 × 2 × 1 = 720

SC-32 Verification Keys: [6] [↶] [x!]

Investigation 2: How many ways can 5 runners finish in 1st, 2nd, and 3rd place? (Compute 5P3 = 5! / 2!).

Expected Answer: 5 × 4 × 3 = 60

SC-32 Verification Keys: [5] [↶] [x!] [2] [↶] [x!] [÷]

Investigation 3: How many 2-person partnerships can be formed from 7 students? (Compute 7C2 = 7! / (2! × 5!)).

Expected Answer: (7 × 6) / 2 = 21

SC-32 Verification Keys: [7] [↶] [x!] [2] [↶] [x!] [÷] [5] [↶] [x!] [÷]

Investigation 4: Find the mean x-bar of the two data values 5 and 15.

Expected Answer: (5 + 15) / 2 = 10

SC-32 Verification Keys: [5] [Σ+] [1] [5] [Σ+] [↷] [$\bar{x}, \bar{y}$]

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.